Published in final edited form as:
Cancer Lett. 2024 November 01; 604: 217267. doi:10.1016/j.canlet.2024.217267.

Localized Ablative Immunotherapy Enhances Antitumor Immunity by Modulating the Transcriptome of Tumor-Infiltrating Gamma Delta T Cells

Kaili Liu[1], **Ashley R. Hoover**[1,2], **Lin Wang**[1], **Yuanhong Sun**[1], **Trisha I. Valerio**[1], **Coline Furrer**[1], **Jacob Adams**[1], **Jingxuan Yang**[3,4], **Min Li**[3,4], **Wei R. Chen**[1,4,*]

[1]Stephenson School of Biomedical Engineering, University of Oklahoma, Norman, OK, USA

[2]Arthritis and Clinical Immunology Research Program, Oklahoma Medical Research Foundation, Oklahoma City, OK, USA

[3]Department of Medicine, Department of Surgery, University of Oklahoma Health Sciences Center, Oklahoma City, OK 73104, USA

[4]Stephenson Cancer Center, University of Oklahoma Health Sciences Center, Oklahoma City, OK 73104, USA

Abstract

Gamma delta T cells (γδT cells) play crucial roles in the immune response against tumors, yet their functional dynamics under different cancer therapies remain poorly understood. Laser Ablative Immunotherapy (LAIT) is a novel cancer treatment modality combining local photothermal therapy (PTT) and intratumoral injection of an immunostimulant, N-dihydrogalactochitosan (glycated chitosan, GC). LAIT has been shown to induce systemic antitumor immune responses in pre-clinical studies and clinical trials, eradicating both treated local tumors and untreated distant metastases. In this study, we used LAIT to treat breast tumors in a mouse model and investigated the effects of LAIT on tumor-infiltrating γδT cells using single-cell RNA sequencing (scRNAseq). We characterized the γδT cells from tumors in control,

[*]Corresponding author: Wei-R-Chen@ou.edu.

CRediT authorship contribution statement
Kaili Liu: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Visualization, Validation, Writing – original draft. Ashley R. Hoover: Investigation, Data curation, Methodology. Lin Wang: Investigation, Writing – review & editing. Yuanhong Sun: Investigation, Data curation. Trisha I. Valerio: Investigation. Coline Furrer: Methodology, Investigation. Jacob Adams: Investigation. Jingxuan Yang: Writing – review & editing. Min Li: Writing – review & editing. Wei R. Chen: Conceptualization, Supervision, Project administration, Methodology, Funding acquisition, Writing – review & editing.

Declaration of competing interest
Wei R. Chen is co-founder and an unpaid member of the Board of Directors of Immunophotonics, Inc. The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and code availability
The scRNAseq data supporting this study is accessible under GEO accession number GSE150675 [23, 52]. Analysis for such data can be available upon request.

PTT, GC, and LAIT (PTT+GC) groups, by identifying six distinct subtypes: activated, cytotoxic, cycling cytotoxic, IFN-enriched, antigen-presenting, and IL17-producing γδT cells. Differential gene expression analysis revealed that LAIT significantly upregulated genes associated with T cell activation, leukocyte adhesion, and interferon signaling in treated tumor tissues while downregulating genes involved in protein folding and stress responses. LAIT also uniquely increased the proportion of IL17-producing γδT cells, which correlated with prolonged survival in breast cancer patients, as analyzed using TCGA data. Furthermore, the transcriptional profiles of γδT cells in LAIT-treated tumors closely resembled those in immune checkpoint inhibitor (ICI)-treated patients, suggesting potential synergistic effects. Our findings indicate that LAIT modulates the γδT cell transcriptome, enhancing their antitumor capabilities and providing a basis for combining LAIT with ICI therapy to improve cancer treatment outcomes.

Keywords

Single-cell RNA sequencing; γδT cell activation; localized ablative immunotherapy; N-dihydrogalactochitosan; breast cancer

Introduction

Gamma delta T cells (γδT cells) represent a distinct subset of T cells that express the γδT cell receptor (TCR) instead of the conventional αβTCR [1]. These cells are pivotal in the immune system's ability to combat tumors and infections, as they recognize antigens independently of major histocompatibility complex (MHC) molecules [2]. This capability allows γδT cells to detect a wide array of antigens, critical for cancer immunotherapy [3–7].

γδT cells serve as a crucial link between innate and adaptive immunities, exhibiting properties that contribute to both immune response types [8]. They can be categorized into different subsets based on their TCR composition and functional roles, with particular attention given to IL17-producing (IL17-) and IFNγ-producing (IFNγ-) γδT cells in the context of cancer immunology [9]. IL17-γδT cells are primarily known for their role in promoting inflammation. Their involvement in cancer is complex, as they have been shown to both support and inhibit tumor growth depending on the environmental context [10]. For instance, IL17-γδT cells can enhance tumor progression by promoting angiogenesis and recruiting neutrophils and macrophages to the tumor site [11, 12]. Some studies also showed that IL17-γδT cells can enhance the recruitment of certain immune cells to the tumor site, potentially augmenting the antitumor immune response [13, 14]. On the other hand, IFNγ-γδT cells are associated with potent antitumor responses, characterized by their cytotoxic activity and ability to activate other immune cells to fight against cancer [15–17].

Localized ablative immunotherapy (LAIT) combines photothermal therapy (PTT)-generated physical destruction of local tumor tissue with intratumoral delivery of an immunostimulant, N-dihydrogalactochitosan (glycated chitosan, GC), to induce a systemic immunomodulation to boost the body's antitumor immune response against both the treated primary tumors and untreated distant metastases [18–21]. Previous studies have shown that LAIT can effectively modulate tumor-infiltrating lymphocytes, including both αβT [22] and B [23] cells. However, the specific impact of LAIT on γδT cells has not been explored. Understanding the response of γδT cells to LAIT could provide critical insights to enhance cancer immunotherapy strategies.

In this study, we utilized single-cell RNA sequencing (scRNAseq) to explore the transcriptional landscape of tumor-infiltrating γδT cells across different treatment modalities. Our specific objectives are: 1) to investigate the gene expression profiles and functional pathways of γδT cells in response to PTT, GC, and LAIT (PTT+GC) treatments in a preclinical breast tumor model; 2) to evaluate the synergistic effects of GC and PTT by comparing their impact on γδT cells with those observed under PTT and GC treatments individually; 3) to identify additional treatment modalities that may exhibit synergistic potential when combined with LAIT; 4) to analyze the correlation between LAIT-induced responses in γδT cells and clinical survival outcomes in breast cancer patients. We found significant transcriptional alterations in γδT cells following LAIT, with notable enhancements in genes associated with T cell activation, leukocyte adhesion, and interferon signaling pathways. Furthermore, the transcriptional profiles of γδT cells in LAIT-treated tumors closely resembled those of immune checkpoint inhibitor (ICI)-treated patients, suggesting potential synergistic effects. Importantly, the expression level of LAIT-induced genes was positively associated with improved survival rates in breast cancer patients. These findings suggest that LAIT enhances the antitumor capabilities of γδT cells, supporting the idea of combining LAIT with other immunotherapeutic approaches, such as ICIs, to improve cancer treatment efficacies.

Results

ScRNAseq reveals the landscape of tumor-infiltrating γδT cells upon LAIT

We previously reported the therapeutic effects of LAIT on MMTV-PyMT breast tumors, highlighting its modulatory roles on tumor-infiltrating T [22] and B cells [23]. While there are numerous studies revealing the atlas and roles of αβT cells (CD4 and CD8) in antitumor functions using scRNAseq[24–26], in this study we focus on the actions of γδT cells, which remain less understood. Following the treatment and scRNAseq analysis workflow (Figure S1A), we subset the γδT cells from 49,380 CD45$^+$ tumor-infiltrating lymphocytes (11,584 CTRL; 14,071 PTT; 12,501 GC; 11,224 LAIT) and identified 4,235 γδT cells (1,683 CTRL; 1,139 PTT; 792 GC; 621 LAIT). Feature plots showed that these cells expressed *Cd3g* and *Tcrg-C1/C4* but lacked *Cd8a/b1*, *Cd4*, *Cd19* or *Itgam* (CD11b), as shown in Figure S1B, ensuring that CD8$^+$ T, CD4$^+$ T, B and myeloid cells were successfully filtered out and only γδT cells were retained for downstream analysis. After integrating single-cell data, re-clustering and annotating cell types [27], six clusters (0-5) were identified using Uniform Manifold Approximation and Projection (UMAP) (Figure S1C). The clusters were further identified as different subset of γδT cells (Figure 1A), based on the common immune markers (Figure 1B–C). Furthermore, we separated the γδT subtypes across the groups treated by different components of LAIT: CTRL, PTT, GC, and PTT+GC (LAIT), as shown in Figure 1D.

We next investigated the constitution and distribution of γδT subtypes across treatment groups by calculating the cell numbers and proportions (Figure 1E, S1D–E). PTT, GC, and LAIT treatments resulted in an increasing trend in the IL17-γδT cell subtype proportion and a decreasing trend in the IFNγ-γδT cell subtypes (Figure S1D–E). This trend was also confirmed by normalizing the γδT subtype proportions to the number of $CD45^+$ tumor-infiltrating immune cells in each treatment group (Figure 1E). For example, IL17-γδT cell proportion induced from 0.4% in CTRL to 1.2% by LAIT, representing a 3-fold increase. Given the reported antitumor roles of IL17-γδT cells, these results suggest that the antitumor functions of LAIT may be associated with inducing the frequencies and transcriptomes of IL17-γδT cells.

Furthermore, by comparing Figure 1A and Figure S1C, cluster 0 was identified as cytotoxic γδT (Cyto.gdT) cells because of the expression of *Gzmb*. Cluster 1 was named as cycling cytotoxic γδT (Cyc.Cyto.gdT) cells due to the expressions of *Gzmb* and *Mki67*. Cluster 2 was characterized as activated γδT cells (Act.gdT) due to the expressions of naive markers *Sell* (CD62L), *Lef1, left*, and activation marker *Cd28* (Figure 1B, S1F). The cells in this cluster also expressed activation marker *Cd69*, effector marker *Cd44* and memory marker *Il7r*, indicating that it can also be phenotypically named as central memory γδT cells (CM.gdT). Cluster 3 was annotated as IL17-expressing γδT cells (Il17.gdT) due to specific expression of *Il17a*; this cluster was also expressed with activation marker *Cd69*, effector marker *Cd44*, memory marker *Il7r* exhaustion marker/immune checkpoint *Pdcd1* (PD1), and *Ctla4* (CTLA4), suggesting that this cluster was also with exhausted and central memory state (Figure 1B, S1F). Interferon response genes, including *Ifit1*, *Irf7*, and *Ifng* were highly expressed in cells in cluster 4, which was identified as IFN.gdT cells. Cluster 5 was characterized as antigen-presenting cell-like γδT cells (APC.gdT) due to specific expressions of major histocompatibility complex (MHC) molecule transcripts *H2-Aa* and *H2-DMa*. Previous reports showed that γδT cells could be classified as IFNγ-expressing and IL17-expressing subtypes [3, 8, 9].

To validate the clustering and annotation of γδT cell subtypes, we performed trajectory inference and found these cell subtypes were localized with distinct and overlapping patterns in the branches of the trajectory (Figure 1F–H). For example, IL17-γδT cells were localized on the rightest terminals of branches, showing the furthest against other subtypes. Activated/central memory γδT cells were distributed mainly on the right branches, while cytotoxic γδT cells were on almost all branches with the cycling cells on the most left side (Figure 1H). Pseudotime-dependent gene expression analysis (Figure 1I) showed that cytotoxic gene *Gzmb* was expressed early in the pseudotime. Conversely, *Il17a*, *Pdcd1*, *Ctla4*, *Il7r*, *Cd44*, and *Cd69* were highly expressed at the terminal pseudotime, where IL17-γδT cells resided. Therefore, This trajectory analysis agreed with the cell subtype annotation visualized by UMAP (Figure S1F).

Differential gene expression and functional enrichment analysis of tumor-infiltrating γδT cells

To explore the effects of each treatment on the functions of tumor-infiltrating γδT cells at the transcriptional level, we conducted analyses for differential gene expression (DGE) and functional enrichment. We performed three comparisons: PTT vs CTRL, GC vs CTRL, and LAIT vs CTRL.

For PTT vs CTRL, 25 genes were upregulated, and 24 were downregulated as shown in the volcano plot (Figure 2A). Biological process (BP) of gene ontology (GO) analysis using the over-representation analysis (ORA) method demonstrated that PTT enriched antigen processing and presentation signaling pathways compared to CTRL (Figure 2D). PTT also enriched IL17 signaling as indicated by KEGG analysis (Figure S2A). Conversely, PTT downregulated genes involved in chaperon-mediated protein folding (Figure 2D), protein processing, splicesome (Figure S2A), and regulation of HSF-1-mediated heat shock response (Figure S2B).

GC stimulation upregulated 123 genes and downregulated 79 genes compared to CTRL (Figure 2B). GC upregulated genes involved in response to virus (Figure 2D), IL17 signaling, T cell receptor (TCR) signaling pathway (Figure S2A), and IFNγ signaling (Figure S2B), suggesting that GC can activate γδT cells and induce an antitumor phenotype. Conversely, GC downregulated genes involved in protein folding/processing (Figure 2D, S2A), spliceosome, and heat shock response (Figure S2A–B).

When comparing LAIT (PTT+GC) vs CTRL, we observed 249 upregulated and 123 downregulated genes (Figure 2C), more than the numbers of differentially expressed genes (DEGs) from PTT and GC treatments. LAIT induced genes involved in T cell activation, leukocyte cell-cell adhesion, and response to virus (Figure 2D), TCR signaling (Figure S2A), and interferon signaling (Figure S2B). LAIT downregulated genes involved in pathways similar to those affected by GC (Figure 2D, Figure S2A–B).

The ORA analysis revealed that PTT, GC, and LAIT induced overlapping gene expression signatures (Figure 2D), although the numbers and fold changes of DEGs were not identical (Figure 2A–C). To further explore gene expression differences induced by these treatments, we used hallmark gene sets of MsigDB (Figure 2E–G), KEGG (Figure S2C), Reactome pathway (Figure S2E) databases and performed gene set enrichment analysis (GSEA) to visualize gene-pathway networks. PTT elevated TNFα signaling pathways (Figure 2E). GC enriched P53, IFNγ and TNFα signaling while downregulating mTORC1 and cellular responses to external stimuli pathways (Figure 2F, S2D). LAIT upregulated IFNγ signaling, TNFα signaling, inflammatory response (Figure 2G), cytokine signaling (Figure S2E), and downregulated Myc targets (Figure 2G), autophagy signaling, and heat shock protein pathways (Figure S2E). These analyses showed that PTT, GC, and LAIT had overlapping pathways including the TNFα responses. However, LAIT enriched pathways with the greatest number of associated genes, suggesting the synergistic capability of PTT+GC in modulating the γδT cell transcriptome. This implies that LAIT may have a broader impact on activating γδT cell gene signatures and potential antitumor functions than PTT or GC alone.

Overlapping of treatment-upregulated genes in γδT cells

To understand the molecular mechanisms stimulated by LAIT, we investigated the relationships among treatment-regulated DEGs, focusing on LAIT-specific DEGs. A Venn diagram displayed the overlap of PTT, GC, and LAIT regulated genes (Figure 3A). Genes uniquely regulated by PTT alone were confined in Set0. PTT-GC-LAIT intersecting genes were confined in Set1; GC-LAIT intersecting genes (excluding Set1) were confined in Set2.

Genes uniquely regulated by GC alone were grouped in Set3. Similarly, Set4 was defined as specific upregulation by LAIT.

Collectively, 12 genes of Set1 were upregulated among PTT, GC, and LAIT (Figure 3A). Individually, PTT, GC, and LAIT upregulated 5 (Set0), 28 (Set3) and 156 (Set4) genes, respectively (Figure 3A). LAIT specifically upregulated genes enriched in the pathways of response to IFNα/β/γ (Figure 3B, S3A), cytokine production, leukocyte cell-cell production (Figure 3B), TCR signaling, and NFκB signaling (Figure S3B). These pathways were shared by enrichment from Set2 with 78 genes, a major overlapping gene set between GC and LAIT (Figure 3A). This result suggests that GC's effect in modulating the gene expression in γδT cells was closer to LAIT than PTT.

We compared the expression levels of each gene set component in different treatment groups using circular heatmap. Set0 was not used for analysis due to the small number of genes. Even though Set1 genes were common to all treatment groups, genes with the highest level of expression were largely found in the LAIT group (Figure 3C). This was also observed in gene Set2 (shared by GC and LAIT) for selected genes (Figure 3D), implying that LAIT has a greater capacity for activating γδT cells than GC alone. Consistent with the pattern established in Set1 and Set2, Set4 genes showed the highest expression on γδT cells in the LAIT treated tumors (Figure 3F). In GC-specific upregulated gene Set3, GC treatment showed the highest upregulation (Figure 3E).

Instead of showing the individual expression of genes from Set1 to Set4 in each treatment, we next used *AddModuleScore* function from *Seurat* R package [27] to calculate the average expression level of these upregulated gene sets. These expression scores for each treatment were displayed using violin plots (Figure S3D–G). The expression scores in gene Set1 (Figure S3D) and Set2 (Figure S3E) for GC and LAIT treatments were higher than that in CTRL and PTT, indicating that both PTT and GC upregulated gene expression in γδT cells compared with CTRL, and that combination of GC with PTT achieved a synergistic effect in modulating the transcriptome. LAIT has the highest gene module score in Set4 because this gene set is LAIT-specifically upregulated (Figure S3G), indicating that PTT and GC can synergize to promote the expression of gene signatures in Set4. This result was consistent with that of the gene expression profiling (Figure 3C–F).

We next investigated the transcriptional changes of representative genes in each γδT cell clusters in responding to LAIT (Figure 3G). *Tcrg-C1* was stimulated by LAIT in Il17-γδT cell cluster, while *Tcrg-C4* was upregulated in almost all γδT cell subtypes. *Cd3g/d/e* showed an increasing expression pattern from treatments of PTT, GC, and LAIT in activated and IL17-γδT cells. *Sell* and *Tcf7* were enriched in activated and IFN-enriched γδT cells, and their expressions were driven by GC and LAIT (Figure 3G). *Cd27* and *Cd28* were most abundantly expressed in activated and IFN-enriched γδT cells. While *Cd27* was slightly stimulated by LAIT in activated γδT cells, both *Cd27* and Cd28 were strongly induced by treatments of PTT, GC and LAIT in IFN-enriched γδT cells. Activation marker *Cd69* was ubiquitously expressed in all γδT cells (Figure 3G) while its upregulation by LAIT occurred in activated, cytotoxic, IFN-enriched and Il17-γδT cells (Figure 3G). Stress responsive gene *Jun*, *Fos*, and *Fosb* were mainly activated by all treatments in all γδT cell subtypes with variable degrees. Effector marker *Cd44, Il2ra*, co-stimulatory molecule *Icos, Zbtb16*, *Runx1* were highly expressed Il17-γδT cells but showed no apparent changes upon treatments. *Il17a* and *Pdcd1* were also highly expressed in this γδT cluster, their expressions were driven by GC and LAIT. Cytotoxic molecule *Gzmb* was highly expressed in cytotoxic, cycling cytotoxic, IFN-enriched, and APC-γδT cells, but its expression showed no apparent affection by treatment. *Prf1* and *Ifng* showed an increasing expression pattern upon treatments of PTT, GC, and LAIT in activated and cytotoxic γδT cells. While *Tnf* exhibited an increasing expression pattern by treatment in all γδT subtypes, *Tnfsf10* (TRAIL) showed smaller response to treatment in (cycling) cytotoxic and APC-γδT cell clusters.

Also shown in Figure 3G, anti-viral pathway component *Ddx58* (RIG-I) and *Ifih1* (MDA5) were mainly expressed and stimulated by GC and LAIT, in IFN-enriched, APC, and IL17-γδT cell subtypes. Their downstream molecules, including transcription factors of *Stat1, Stat4, Stat6, Irf1, Irf2, Irf7*, interferon stimulated and induced genes such as *Isg15, Mx1, Ifi47, Ifi206, Ifi208, Ifi209, Ifit1, Ifit2*, and *Ifit3*, showed similar regulation patterns and were markedly stimulated by GC and LAIT in IFN-enriched, APC and IL17-γδT cells. Proinflammatory genes *Il1b*, and *S100a9* were highly expressed in APC γδT cells and induced by LAIT in all γδT cell subtypes. These results showed that combination therapy LAIT enabled T cell activation and stimulation of the interferon-related pathways in γδT cells.

Overlapping of treatment-downregulated genes in γδT cells

We next investigated the relationship between treatment-downregulated DEGs, with a focus on the LAIT-specific negative regulation on these genes in γδT cells (Figure 4A). Genes intersected with different treatments were grouped similarly to Figure 3A. Collectively, for Set1, PTT, GC and LAIT commonly downregulated 18 genes (Figure 4A). GC and LAIT (Set2) co-downregulated 51 genes, while individually, PTT downregulated 1 gene (Set0), GC downregulated 9 genes (Set3), and LAIT downregulated 50 genes (Figure 4A). Signaling pathways including chaperon-mediated protein folding/processing (Figure 4B, S4B), mTORC1/Myc targets (Figure S4A), cell responses to heat shock/stress/external stimuli (Figure S4C) were found among Sets1 to 4.

We compared the expression levels of these confined gene sets in different treatment groups using circular heatmap analysis. The gene expression pattern showed an increasing inhibition from CTRL to PTT, GC and LAIT in Sets 1, 2 and 4, suggesting the unique and combinatory effect of each treatment in regulating these gene expression profiles (Figure 4C–F). In Set3, GC showed the highest inhibition due to these genes being GC-specific downregulated genes (Figure 4E).

Next, we calculated the average expression level of these downregulated genes from Set1 to Set4 in each treatment (Figure S4D–G). The expression scores for gene Set1 (Figure S4D) and Set2 (Figure S4E) were lower in GC and LAIT than that in CTRL and PTT. GC showed the lowest downregulation on expression score in Set3 (Figure S4F) while LAIT exhibited the lowest score in Set4 (Figure S4G).

We then detected the selected downregulated genes in each treatment in each γδT cell subtype (Figure 4G). LAIT downregulated expressions of a series of immune-suppressive molecules, including *Foxo1, Irf2bp2, Nt5e* (CD73), *Tgfb1*, and *Vegfa* in almost all γδT cells with variable degrees. Another panel of heat shock proteins, including *Hspb1, Hsph1, Hspd1, Hspe1, Hsp90aa1, Dnajb1, Hspa1a*, and *Hspa1b* were also downregulated by treatments, especially LAIT, across all the γδT subpopulations (Figure 4G).

Correlation between LAIT and ICI in stimulating antitumor phenotype of γδT cells

We utilized the published antitumor/pro-tumor gene signatures that have been reported in human breast cancer [28] and detected their module expression levels in γδT cells from the mouse tumor microenvironment (TME) in response to LAIT treatment in our study. PTT, GC, and LAIT all stimulated antitumor gene signatures, including T cell activation, pro-inflammatory function, Type I/II IFNs, and inhibited the antitumor signature (Figure 5A). GC drove the expression alternations of these gene signatures, resulting in the highest module scores in antitumor and the lowest in anti-inflammatory/pro-tumor gene modules in the GC and LAIT treatment groups. These results suggested that LAIT enabled upregulation of type I/II IFN and pro-inflammatory pathways and contribute to the induction of cytotoxic tumor killing antitumor phenotypes in γδT cells.

ICI has achieved significant progress in cancer treatment. However, its effect on γδT cells are not well understood [29, 30]. In this study, we extracted from human breast cancer patients the γδT cell scRNAseq data, with subtypes annotated as 'γδT' and 'Vg9Vd2 γδT' cells (Figure S5A), as well as the scRNAseq data before and after ICI treatment (Figure S5B). The quality control was performed (Figure S5C) similar to the analysis for our PyMT γδT cells (Figure S1B). We found that ICI therapy induced elevated expression patterns of antitumor gene signatures including T cell activation, pro-inflammatory, type I and II interferon (Figure 5B), consistent with our results of GC and LAIT treatments (Figure 5A). Conversely, ICI downregulated anti-inflammatory signatures, leading to elevated pro-inflammatory functions (Figure 5B), corresponding to the results in Figure 5A. These results suggested that ICI could induce γδT cell antitumor activation and cytotoxicity, resembling the effects of LAIT .

Based on the similar expression patterns of γδT cell antitumor gene signatures between LAIT and ICI, we further explored a more direct relation between their effects by 1) determining the LAIT-induced DEGs, focusing on the upregulated genes in γδT cells from ICI-treated breast cancer patients; 2) determining the ICI-induced DEGs, focusing on the upregulated genes in γδT cells from LAIT-treated mice. ICI-upregulated gene signatures in γδT cells from breast cancer patients had higher expression in PTT-, GC-, and LAIT-treated mouse tumors (Figure 5C). Likewise, LAIT-upregulated gene signatures in γδT cells from the PyMT tumor bearing mice also had higher expression in the tumors of ICI-treated patients (Figure 5D). This correlation analysis confirmed the similarity of modulating transcriptome between LAIT and ICI in γδT cells.

Next, we investigated the similarities between LAIT and ICI by performing enrichment analysis and comparing the induced pathways. Similar to LAIT's modulation effect (Figure 2G), GSEA analysis showed that ICI induced genes enriched in IFNγ response, TNFα signaling, P53 pathway, and UV response up (Figure S5D), further validating the similarity between LAIT and ICI in modulating γδT cells at transcriptional level. GO and KEGG analysis showed that ICI induced genes were enriched in TCR signaling, response to TNF, T cell activation/differentiation, and PD-1/PD-L1 pathway (Figure S5E). Taken together, these results further confirmed the similarity and correlation in modulating γδT cell gene expression in TME by LAIT and ICI, suggesting that the combinations of LAIT and ICI has the potential to provide synergy in exerting antitumor immunity by activating γδT cells.

LAIT-upregulated genes in IL17-γδT cells positively correlate with overall survival of breast cancer patients

To assess the clinical relevance of LAIT's effects on transcriptional modulations, we investigated the potential associations between LAIT-mediated gene expressions in γδT cells and the overall survival of breast cancer patients in the Cancer Genome Atlas (TCGA). To emphasize the synergistic effects of PTT and GC, we focused on the LAIT vs GC-derived upregulated and downregulated genes in IL17-, IFNγ-, and all γδT cells to obtain the enrichment score (ES) based on gene set variation analysis (GSVA) [31] (Figure 6A–C). Breast cancer patients with a higher expression of LAIT-upregulated genes in IL17-γδT cells (Figure 6D), but not in all γδT cells (Figure S6A) or all IFNγ-γδT cells (Figure S6C), exhibited prolonged survival compared to the patients with lower gene expression. This demonstrated that the genes specifically induced by LAIT, but not GC alone, may confer a favorable effect on cancer patient prognosis and survival extension. On the other hand, the LAIT vs GC-derived downregulated genes, in either IL17-γδT cells or IFNγ-γδT or all γδT cells, did not affect patient survival (Figure 6E, S6B, S6D). Taken together, we provided predictive evidence that LAIT treatment modalities have a greater potential to prolong patient survival by driving the specific gene expressions in IL17-γδT cells.

Discussion

Our study elucidates the impact of LAIT on tumor-infiltrating γδT cells in a murine breast cancer model, revealing significant transcriptional changes and functional reprogramming within these cells. The insights garnered from our scRNAseq analysis provide a comprehensive understanding of the dynamic roles that γδT cells play in the TME under various therapeutic conditions, deepening our knowledge of the tumor immune landscape and expanding potential targets for future therapeutic and diagnostic treatments [32].

γδT cells are renowned for their ability to mediate rapid immune responses due to their independence from MHC-restricted antigen presentation [1, 2]. This unique property allows them to recognize a broad spectrum of tumor antigens and respond to cellular stress signals. Our study identified six γδT cell subtypes within the TME: activated, cytotoxic, cycling cytotoxic, IFN-producing, antigen-presenting, and IL17-producing cells. This classification aligns with previous research highlighting the functional heterogeneity of γδT cells and their distinct roles in tumor immunity [3, 33, 34].

The differential gene expression analysis revealed that LAIT significantly upregulates genes associated with T cell activation, leukocyte adhesion, and interferon signaling. Specifically, LAIT treatment increased the proportion of IL17-γδT cells, which is consistent with reports that IL17-γδT cells can promote both pro-inflammatory and antitumor responses, depending on the tumor context [9, 13, 14]. Our findings align with these studies showing that IL17-γδT cells can enhance immune cell recruitment to the tumor site, thereby augmenting the overall antitumor response. Our observations of increased IL17-γδT cells following LAIT treatment are supported by evidence that these cells can exhibit antitumor activity in various cancers. For instance, high densities of IL17-producing cells have been associated with improved prognosis in ovarian cancer [35] and esophageal squamous cell carcinoma [14]. Conversely, other studies suggest that IL17-γδT cells may promote tumor growth in certain contexts, such as colon and gallbladder cancers [11, 36]. This duality underscores the need for context-specific strategies when targeting IL17-γδT cells in cancer therapy. However, the precise factors within the tumor environment that determine the anti- or pro-tumor role of IL17-γδT cells remain poorly understood, both in previous studies and in our work. Additionally, it is unclear whether the expansion of this γδT cell subtype and its antitumoral function can be achieved across other tumor models following LAIT treatment, which requires further research in the future.

IFNγ-γδT cells have been widely recognized for their potent cytotoxic effects and their ability to enhance antitumor immunity through the activation of other immune cells. In our study, although a decreasing trend was observed in the numbers of IFNγ-γδT cells (Figure 1E), which might seem contradictory to their known antitumor role, the expression of some genes relevant to different functions was either upregulated (Figure 3G) or downregulated (Figure 4G), showing the delicate modulation functionality of LAIT on IFNγ-γδT cells. However, the relationships between the decreased cell numbers, altered gene expression profiles, and their contribution to survival rates remain unclear in this study. Taken together, our study demonstrates that LAIT not only promotes the expansion of IL17-γδT cells but also modulates IFNγ-γδT cells, resulting in an overall enhancement of the antitumor immune response.

Additionally, the effects of LAIT on γδT cells are not merely the addition of PTT and GC, but rather a synergistic interaction between the two treatments. This is evidenced by the non-additive increase in the proportion of IL17- γδT cells (Figure 1E) and the presence of specific DEGs in the LAIT group (Figure 3A and 4A). Compared to the CTRL group, the proportion of IL17-γδT cells increased by 0.1% in the PTT group, 0.2% in the GC group, and 0.8% in the LAIT group (Figure 1E). From the perspective of DEGs, LAIT uniquely upregulated 156 genes and downregulated 50 genes (Figure 3A and 4A), none of which overlap with those altered by PTT or GC alone. These findings indicate that LAIT induced synergistic effects on γδT cell subtype expansion and activated new response pathways in γδT cells that are not achieved by PTT or GC individually.

The transcriptional similarities between LAIT-treated γδT cells and those from ICI-treated patients highlight the potential for combining these therapies. ICIs have revolutionized cancer treatment by reactivating exhausted T cells, and our data suggest that LAIT could further potentiate this effect by enhancing γδT cell responses [37, 38]. γδT cells, known for their rapid immune responses and ability to recognize a broad spectrum of tumor antigens, often become dysfunctional within the immunosuppressive TME due to the upregulation of immune checkpoint molecules such as PD-1, CTLA-4, and TIM-3 [39]. Targeting these immune checkpoints has been shown to restore γδT cell function, leading to increased proliferation, activation, and cytotoxicity, consistent with our results (Figure 5B, S5D–E). Our findings suggest that the combination of LAIT and ICIs could create a synergistic antitumor response by not only reactivating exhausted αβT cells but also by further enhancing the antitumor potential of γδT cells. Specifically, LAIT appears to amplify the cytotoxic and cytokine-secreting functions of γδT cells, which are crucial for effective tumor clearance. By simultaneously targeting multiple aspects of the immune response, this combination approach has the potential to induce a more durable and comprehensive antitumor immunity, particularly against tumors that are resistant to monotherapy with ICIs [39, 40]. This synergy between LAIT and ICIs could overcome the limitations of each therapy alone and provide a more robust antitumor strategy [41].

Our study's clinical relevance is underscored by the association between LAIT-induced gene expression changes in γδT cells and improved patient survival. These findings suggest that specific transcriptional profiles could serve as biomarkers for predicting therapeutic outcomes and tailoring immunotherapies to enhance efficacy.

Conclusion

In conclusion, our research provides novel insights into the modulation of γδT cells by LAIT, highlighting its potential to enhance antitumor immunity. Through scRNAseq analysis, we identified six γδT cell subtypes and found that LAIT upregulates genes associated with T cell activation, leukocyte adhesion, and interferon signaling. Importantly, LAIT increased IL17-producing γδT cells, correlating with improved survival in breast cancer patients. In addition, the transcriptomic profiles of γδT cells in LAIT-treated tumors closely resembled those in ICI-treated patients, suggesting a potential synergy between LAIT and ICIs. These findings highlight LAIT's potential to enhance γδT cell-mediated antitumor responses and support its combination with other immunotherapies for improved cancer treatment outcomes. However, the study has limitations, including an unclear understanding of what determines the anti- or pro-tumor roles of IL17-γδT cells and whether their expansion after LAIT treatment is consistent across other tumor models. Additionally, the study does not fully explain the relationship between the decreased numbers of IFNγ-γδT cells, their altered gene expression, and their impact on survival. Future research should aim to clarify these uncertainties, as well as explore the clinical translation of these findings, particularly the integration of LAIT with ICIs, to maximize therapeutic benefits for cancer patients.

Materials and Methods

Study approval

Experimental procedures and handling of mice were performed in accordance with the University of Oklahoma Health Sciences Center (OUHSC) and Oklahoma Medical Research Foundation (OMRF) Institutional Animal Care and Use Committee (IACUC) regulations and approved protocols.

Mice

Female FVB/NJ wild-type (stock #001800) and FVB/N transgenic MMTV-PyMT (stock #002374) mice were purchased from Jackson Laboratories and housed according to institutional guidelines at the University of Oklahoma Health Sciences Center. All studies were approved and adhered to OUHSC Institutional Animal Care and Use Committee protocols (#17-032-HCL).

Sample preparation and single-cell RNA sequencing analysis for γδT cells

Following protocols established in our preceding studies [22, 23], we treated the mice, processed tumor samples, isolated tumor-infiltrating immune cells, and prepared single-cell RNA sequencing libraries, utilizing an Illumina NovaSeq 6000 for paired-end sequencing. The *Cell Ranger* pipeline alongside *Seurat* facilitated our scRNAseq analysis, as detailed previously [23, 27, 42–44]. After quality control, normalization, and integration of total $CD45^+$ immune cells, we excluded $CD8^+$ T, $CD4^+$ T, B and myeloid cells based on specific marker expressions (T cell: *Cd3d, Cd3g, Cd3e, Cd8a, Cd8b1, Cd4*, B cell: *Cd19*, Myeloid cell: *Itgam)*, keeping γδT cells via *Cd3d/g/e* and *Tcrg-C1/C4* markers. Subsequent re-clustering of these γδT cells was performed at a resolution of 0.3 to produce UMAP visualizations, with the *FeaturePlot* function in *Seurat* illustrating selected gene expressions. For single-cell trajectory inference, we employed the *Monocle2* R package [45].

Differential gene expression (DGE) and functional enrichment analysis

We conducted differential gene expression analyses comparing treatments (PTT, GC, LAIT) with control (CTRL), employing the *FindMarkers* function in *Seurat* with default settings [27]. Volcano plots, created using *ggplot2* in *R*, highlighted the top 10 upregulated and downregulated DEGs. For functional enrichment analysis, we utilized the *clusterProfiler* R package, applying the over-representation analysis (ORA) method with GO, KEGG, Reactome, and MsigDB databases [46].

Overlap analysis of treatment-derived DEGs

Treatment-derived DEGs were overlapped and visualized by using *VennDiagram* R package [46]. Expression of these subset genes were obtained using *AverageExpression* function in *Seurat* R package[27] and visualized using circular heatmap [47, 48].

Gene signature selection and module score calculation

Adhering to descriptions in the literature [28], we chose gene signatures indicative of T cell activation, pro-/anti-inflammatory functions, type I/II interferon responses. The *AddModuleScore* function in *Seurat* calculated gene expression module scores for each treatment group, visualized via violin plots produced by *ggpubr*.

ScRNAseq analysis for γδT cell transcriptomics from ICI-treated breast cancer patients

To elucidate the impact of ICI therapy on γδT cells in breast cancer patients, we analyzed the scRNAseq data from anti-PD1 treated patients [49]. According to the author's annotated metadata, the "Pre" labeled patients refers to non-ICI treatments and "On" refers to ICI treatment. Both labelled cell subtypes of ‘gdT’ and ‘Vg9Vd2_gdT’ were selected as γδT cells. 1156 γδT cells from patients were used for downstream analysis.

Correlation between clinical survival and expression of given gene set

We explored the clinical relevance of LAIT-modulated gene expressions using RNAseq and survival data from TCGA datasets for breast cancer. (https://xenabrowser.net/datapages/) [50]. LAIT vs GC comparison was used to obtain the upregulated and downregulated gene sets. Gene symbols in mouse were converted to gene symbols in human by using *biomaRt* R package [51]. By employing gene set variation analysis (GSVA) via the R package GSVA, we correlated gene expression profiles with patient survival outcomes, utilizing Kaplan-Meier estimates and log-rank tests for statistical analysis. Cancer patients were stratified as the “high” group if ES was above the indicated cutoff (such as median) and as “low” group if ES is below the cutoff. The numbers below plots represent numbers of cancer patients and survival time are days.

Statistical analyses

Statistical analyses were conducted in *R*, employing the *prop.test* function for cell proportion tests and Kruskal-Wallis and Wilcoxon tests for multiple and two-group comparisons, respectively, via the *ggpubr* package. Survival differences were assessed using Kaplan-Meier analysis, with adjusted P values $\leqslant 0.05$ indicating statistical significance.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

Acknowledgments

This work was supported in part by the National Cancer Institute (R01CA205348 and R01CA269897), National Institute of General Medical Sciences (P20GM135009), and the Oklahoma Center for the Advancement of Science and Technology (HF20-019 and HR23-069). We would like to thank the Flow Cytometry Core Facility and Genomics Core Facility at the Oklahoma Medical Research Foundation (OMRF) for providing flow cytometry services and assisting in scRNAseq experiment.

References

[1]. Mensurado S, Blanco-Domínguez R, Silva-Santos B, The emerging roles of γδ T cells in cancer immunotherapy, Nat Rev Clin Oncol, 20 (2023) 178–191. [PubMed: 36624304]

[2]. Melandri D, Zlatareva I, Chaleil RAG, Dart RJ, Chancellor A, Nussbaumer O, Polyakova O, Roberts NA, Wesch D, Kabelitz D, Irving PM, John S, Mansour S, Bates PA, Vantourout P, Hayday AC, The γδTCR combines innate immunity with adaptive immunity by utilizing spatially distinct regions for agonist selection and antigen responsiveness, Nat Immunol, 19 (2018) 1352–1365. [PubMed: 30420626]

[3]. Hu Y, Hu Q, Li Y, Lu L, Xiang Z, Yin Z, Kabelitz D, Wu Y, γδ T cells: origin and fate, subsets, diseases and immunotherapy, Signal Transduction and Targeted Therapy, 8 (2023) 434. [PubMed: 37989744]

[4]. Choi H, Kim T-G, Jeun S-S, Ahn S, Human gamma-delta (γδ) T cell therapy for glioblastoma: A novel alternative to overcome challenges of adoptive immune cell therapy, Cancer Letters, 571 (2023) 216335. [PubMed: 37544475]

[5]. Zhang C, Zhang C, Wang H, Immune-checkpoint inhibitor resistance in cancer treatment: Current progress and future directions, Cancer Letters, 562 (2023) 216182. [PubMed: 37076040]

[6]. Wang Q, Qin Y, Li B, CD8+ T cell exhaustion and cancer immunotherapy, Cancer Letters, 559 (2023) 216043. [PubMed: 36584935]

[7]. Lin Q, Wang X, Hu Y, The opportunities and challenges in immunotherapy: Insights from the regulation of PD-L1 in cancer cells, Cancer Letters, 569 (2023) 216318. [PubMed: 37454966]

[8]. Lo Presti E, Dieli F, Fourniè JJ, Meraviglia S, Deciphering human γδ T cell response in cancer: Lessons from tumor-infiltrating γδ T cells, Immunological Reviews, 298 (2020) 153–164. [PubMed: 32691450]

[9]. Park JH, Lee HK, Function of γδ T cells in tumor immunology and their application to cancer therapy, Experimental & Molecular Medicine, 53 (2021) 318–327. [PubMed: 33707742]

[10]. Deng J, Yin H, Gamma delta (γδ) T cells in cancer immunotherapy; where it comes from, where it will go?, European Journal of Pharmacology, 919 (2022) 174803. [PubMed: 35131312]

[11]. Wu P, Wu D, Ni C, Ye J, Chen W, Hu G, Wang Z, Wang C, Zhang Z, Xia W, Chen Z, Wang K, Zhang T, Xu J, Han Y, Zhang T, Wu X, Wang J, Gong W, Zheng S, Qiu F, Yan J, Huang J, γδT17 cells promote the accumulation and expansion of myeloid-derived suppressor cells in human colorectal cancer, Immunity, 40 (2014) 785–800. [PubMed: 24816404]

[12]. Ma S, Cheng Q, Cai Y, Gong H, Wu Y, Yu X, Shi L, Wu D, Dong C, Liu H, IL-17A produced by γδ T cells promotes tumor growth in hepatocellular carcinoma, Cancer Res, 74 (2014) 1969–1982. [PubMed: 24525743]

[13]. Takeuchi A, Dejima T, Yamada H, Shibata K, Nakamura R, Eto M, Nakatani T, Naito S, Yoshikai Y, IL-17 production by γδ T cells is important for the antitumor effect of Mycobacterium bovis bacillus Calmette-Guérin treatment against bladder cancer, European Journal of Immunology, 41 (2011) 246–251. [PubMed: 21182095]

[14]. Lv L, Pan K, Li X.-d., She K.-l., Zhao J.-j., Wang W, Chen J.-g., Chen Y.-b., Yun J.-p., Xia J.-c., The Accumulation and Prognosis Value of Tumor Infiltrating IL-17 Producing Cells in Esophageal Squamous Cell Carcinoma, PLOS ONE, 6 (2011) e18219. [PubMed: 21483813]

[15]. Gibbons DL, Haque SFY, Silberzahn T, Hamilton K, Langford C, Ellis P, Carr R, Hayday AC, Neonates harbour highly active γδ T cells with selective impairments in preterm infants, European Journal of Immunology, 39 (2009) 1794–1806. [PubMed: 19544311]

[16]. Li Y, Li G, Zhang J, Wu X, Chen X, The Dual Roles of Human γδ T Cells: Anti-Tumor or Tumor-Promoting, Frontiers in Immunology, 11 (2021).

[17]. He W, Hao J, Dong S, Gao Y, Tao J, Chi H, Flavell R, O'Brien RL, Born WK, Craft J, Han J, Wang P, Zhao L, Wu J, Yin Z, Naturally Activated Vγ4 γδ T Cells Play a Protective Role in Tumor Immunity through Expression of Eomesodermin, The Journal of Immunology, 185 (2010) 126–133. [PubMed: 20525896]

[18]. Wang M, Rao J, Wang M, Li X, Liu K, Naylor MF, Nordquist RE, Chen WR, Zhou F, Cancer photo-immunotherapy: from bench to bedside, Theranostics, 11 (2021) 2218–2231. [PubMed: 33500721]

[19]. Korbelik M, Banáth J, Zhang W, Gallagher P, Hode T, Lam SSK, Chen WR, N-dihydrogalactochitosan as immune and direct antitumor agent amplifying the effects of photodynamic therapy and photodynamic therapy-generated vaccines, International Immunopharmacology, 75 (2019) 105764. [PubMed: 31352327]

[20]. Chen WR, Singhal AK, Liu H, Nordquist RE, Antitumor immunity induced by laser immunotherapy and its adoptive transfer, Cancer Res, 61 (2001) 459–461. [PubMed: 11212231]

[21]. Chen WR, Liu H, Ritchey JW, Bartels KE, Lucroy MD, Nordquist RE, Effect of different components of laser immunotherapy in treatment of metastatic tumors in rats, Cancer Res, 62 (2002) 4295–4299. [PubMed: 12154032]

[22]. Hoover AR, Liu K, DeVette CA, Krawic JR, Medcalf AD, West CL, Hode T, Lam SS, Welm AL, Sun XH, Single-cell RNA sequencing reveals localized tumour ablation and intratumoural immunostimulant delivery potentiate T cell mediated tumour killing, Clinical and Translational Medicine, 12 (2022) e937. [PubMed: 35808806]

[23]. Liu K, Hoover AR, Krawic JR, DeVette CI, Sun XH, Hildebrand WH, Lang ML, Axtell RC, Chen WR, Antigen presentation and interferon signatures in B cells driven by localized ablative cancer immunotherapy correlate with extended survival, Theranostics, 12 (2022) 639–656. [PubMed: 34976205]

[24]. Zhang Y, Zuo C, Li Y, Liu L, Yang B, Xia J, Cui J, Xu K, Wu X, Gong W, Liu Y, Single-cell characterization of infiltrating T cells identifies novel targets for gallbladder cancer immunotherapy, Cancer Letters, 586 (2024) 216675. [PubMed: 38280478]

[25]. Du Y, Cai Y, Lv Y, Zhang L, Yang H, Liu Q, Hong M, Teng Y, Tang W, Ma R, Wu J, Wu J, Wang Q, Chen H, Li K, Feng J, Single-cell RNA sequencing unveils the communications between malignant T and myeloid cells contributing to tumor growth and immunosuppression in cutaneous T-cell lymphoma, Cancer Letters, 551 (2022) 215972. [PubMed: 36265653]

[26]. Wang Z, Zhao Y, Wo Y, Peng Y, Hu W, Wu Z, Liu P, Shang Y, Liu C, Chen X, Huang K, Chen Y, Hong H, Li F, Sun Y, The single cell immunogenomic landscape after neoadjuvant immunotherapy combined chemotherapy in esophageal squamous cell carcinoma, Cancer Letters, 593 (2024) 216951. [PubMed: 38734159]

[27]. Butler A, Hoffman P, Smibert P, Papalexi E, Satija R, Integrating single-cell transcriptomic data across different conditions, technologies, and species, Nat Biotechnol, 36 (2018) 411–420. [PubMed: 29608179]

[28]. Azizi E, Carr AJ, Plitas G, Cornish AE, Konopacki C, Prabhakaran S, Nainys J, Wu K, Kiseliovas V, Setty M, Choi K, Fromme RM, Dao P, McKenney PT, Wasti RC, Kadaveru K, Mazutis L, Rudensky AY, Pe'er D, Single-Cell Map of Diverse Immune Phenotypes in the Breast Tumor Microenvironment, Cell, 174 (2018) 1293–1308.el236. [PubMed: 29961579]

[29]. Dougan M, Dranoff G, Dougan SK, Cancer Immunotherapy: Beyond Checkpoint Blockade, Annual Review of Cancer Biology, 3 (2019) 55–75.

[30]. Bagchi S, Yuan R, Engleman EG, Immune Checkpoint Inhibitors for the Treatment of Cancer: Clinical Impact and Mechanisms of Response and Resistance, Annual Review of Pathology: Mechanisms of Disease, 16 (2021) 223–249.

[31]. Hänzelmann S, Castelo R, Guinney J, GSVA: gene set variation analysis for microarray and RNA-Seq data, BMC Bioinformatics, 14 (2013) 7. [PubMed: 23323831]

[32]. Siminzar P, Tohidkia MR, Eppard E, Vahidfar N, Tarighatnia A, Aghanejad A, Recent Trends in Diagnostic Biomarkers of Tumor Microenvironment, Mol Imaging Biol, 25 (2023) 464–482. [PubMed: 36517729]

[33]. Carding SR, Egan PJ, γδ T cells: functional plasticity and heterogeneity, Nature Reviews Immunology, 2 (2002) 336–345.

[34]. Li Z, Yang Q, Tang X, Chen Y, Wang S, Qi X, Zhang Y, Liu Z, Luo J, Liu H, Ba Y, Guo L, Wu B, Huang F, Cao G, Yin Z, Single-cell RNA-seq and chromatin accessibility profiling decipher the heterogeneity of mouse γδ T cells, Science Bulletin, 67 (2022) 408–426. [PubMed: 36546093]

[35]. Lan C, Huang X, Lin S, Huang H, Cai Q, Lu J, Liu J, High density of IL-17-producing cells is associated with improved prognosis for advanced epithelial ovarian cancer, Cell and Tissue Research, 352 (2013) 351–359. [PubMed: 23397428]

[36]. Patil RS, Shah SU, Shrikhande SV, Goel M, Dikshit RP, Chiplunkar SV, IL17 producing γδT cells induce angiogenesis and are associated with poor survival in gallbladder cancer patients, International Journal of Cancer, 139 (2016) 869–881. [PubMed: 27062572]

[37]. Sanmamed MF, Chen L, A Paradigm Shift in Cancer Immunotherapy: From Enhancement to Normalization, Cell, 175 (2018) 313–326. [PubMed: 30290139]

[38]. Kyrysyuk O, Wucherpfennig KW, Designing Cancer Immunotherapies That Engage T Cells and NK Cells, Annual Review of Immunology, 41 (2023) 17–38.

[39]. Gao Z, Bai Y, Lin A, Jiang A, Zhou C, Cheng Q, Liu Z, Chen X, Zhang J, Luo P, Gamma delta T-cell-based immune checkpoint therapy: attractive candidate for antitumor treatment, Molecular Cancer, 22 (2023) 31. [PubMed: 36793048]

[40]. Nada MH, Wang H, Hussein AJ, Tanaka Y, Morita CT, PD-1 checkpoint blockade enhances adoptive immunotherapy by human Vγ2Vδ2 T cells against human prostate cancer, Oncolmmunology, 10 (2021) 1989789.

[41]. Kabelitz D, Serrano R, Kouakanou L, Peters C, Kalyan S, Cancer immunotherapy with γδ T cells: many paths ahead of us, Cellular & Molecular Immunology, 17 (2020) 925–939. [PubMed: 32699351]

[42]. Stuart T, Butler A, Hoffman P, Hafemeister C, Papalexi E, Mauck WM 3rd, Hao Y, Stoeckius M, Smibert P, Satija R, Comprehensive Integration of Single-Cell Data, Cell, 177 (2019) 1888–1902 e1821. [PubMed: 31178118]

[43]. Bajana S, Pankow A, Liu K, Michniowska M, Urban JF, Chen WR, Sun X-H, Correlation between circulating innate lymphoid cell precursors and thymic function, iScience, 25 (2022) 103732. [PubMed: 35118353]

[44]. Hao Y, Hao S, Andersen-Nissen E, Mauck WM, Zheng S, Butler A, Lee MJ, Wilk AJ, Darby C, Zager M, Hoffman P, Stoeckius M, Papalexi E, Mimitou EP, Jain J, Srivastava A, Stuart T, Fleming LM, Yeung B, Rogers AJ, McElrath JM, Blish CA, Gottardo R, Smibert P, Satija R, Integrated analysis of multimodal single-cell data, Cell, 184 (2021) 3573–3587.e3529. [PubMed: 34062119]

[45]. Qiu X, Mao Q, Tang Y, Wang L, Chawla R, Pliner HA, Trapnell C, Reversed graph embedding resolves complex single-cell trajectories, Nature methods, 14 (2017) 979–982. [PubMed: 28825705]

[46]. Yu G, Wang LG, Han Y, He QY, clusterProfiler: an R package for comparing biological themes among gene clusters, Omics : a journal of integrative biology, 16 (2012) 284–287. [PubMed: 22455463]

[47]. Gu Z, Gu L, Eils R, Schlesner M, Brors B, circlize Implements and enhances circular visualization in R, Bioinformatics, 30 (2014) 2811–2812. [PubMed: 24930139]

[48]. Gu Z, Eils R, Schlesner M, Complex heatmaps reveal patterns and correlations in multidimensional genomic data, Bioinformatics, 32 (2016) 2847–2849. [PubMed: 27207943]

[49]. Bassez A, Vos H, Van Dyck L, Floris G, Arijs I, Desmedt C, Boeckx B, Vanden Bempt M, Nevelsteen I, Lambein K, Punie K, Neven P, Garg AD, Wildiers H, Qian J, Smeets A, Lambrechts D, A single-cell map of intratumoral changes during anti-PD1 treatment of patients with breast cancer, Nature Medicine, 27 (2021) 820–832.

[50]. Goldman MJ, Craft B, Hastie M, Repečka K, McDade F, Kamath A, Banerjee A, Luo Y, Rogers D, Brooks AN, Zhu J, Haussler D, Visualizing and interpreting cancer genomics data via the Xena platform, Nature Biotechnology, 38 (2020) 675–678.

[51]. Durinck S, Spellman PT, Birney E, Huber W, Mapping identifiers for the integration of genomic datasets with the R/Bioconductor package biomaRt, Nat Protoc, 4 (2009) 1184–1191. [PubMed: 19617889]

[52]. Hoover AR, Liu K, DeVette CA, Krawic JR, Medcalf AD, West CL, Hode T, Lam SSK, Welm AL, Sun X-H, Hildebrand WH, Chen WR, Single-cell RNA sequencing reveals localized tumour ablation and intratumoural immunostimulant delivery potentiate T cell mediated tumour killing, Clinical and Translational Medicine, 12 (2022) e937. [PubMed: 35808806]

Highlights:

Comprehensive scRNAseq Analysis:

We conducted a detailed single-cell RNA sequencing analysis of tumor-infiltrating γδT cells in a murine breast cancer model under the treatment of different components of LAIT.

γδT Cell Subtype Identification:

We determined six distinct γδT cell subtypes, including activated, cytotoxic, cycling cytotoxic, IFN-producing, antigen-presenting, and IL-17-producing γδT cells.

Impact of LAIT on γδT Cells:

We demonstrated that LAIT significantly enhances T cell activation, leukocyte adhesion, and interferon signaling pathways in γδT cells.

Potential Synergy with ICIs:

We identified the transcriptional similarities between γδT cells in LAIT-treated tumors and that in ICI-treated patients, suggesting potential synergistic effects.

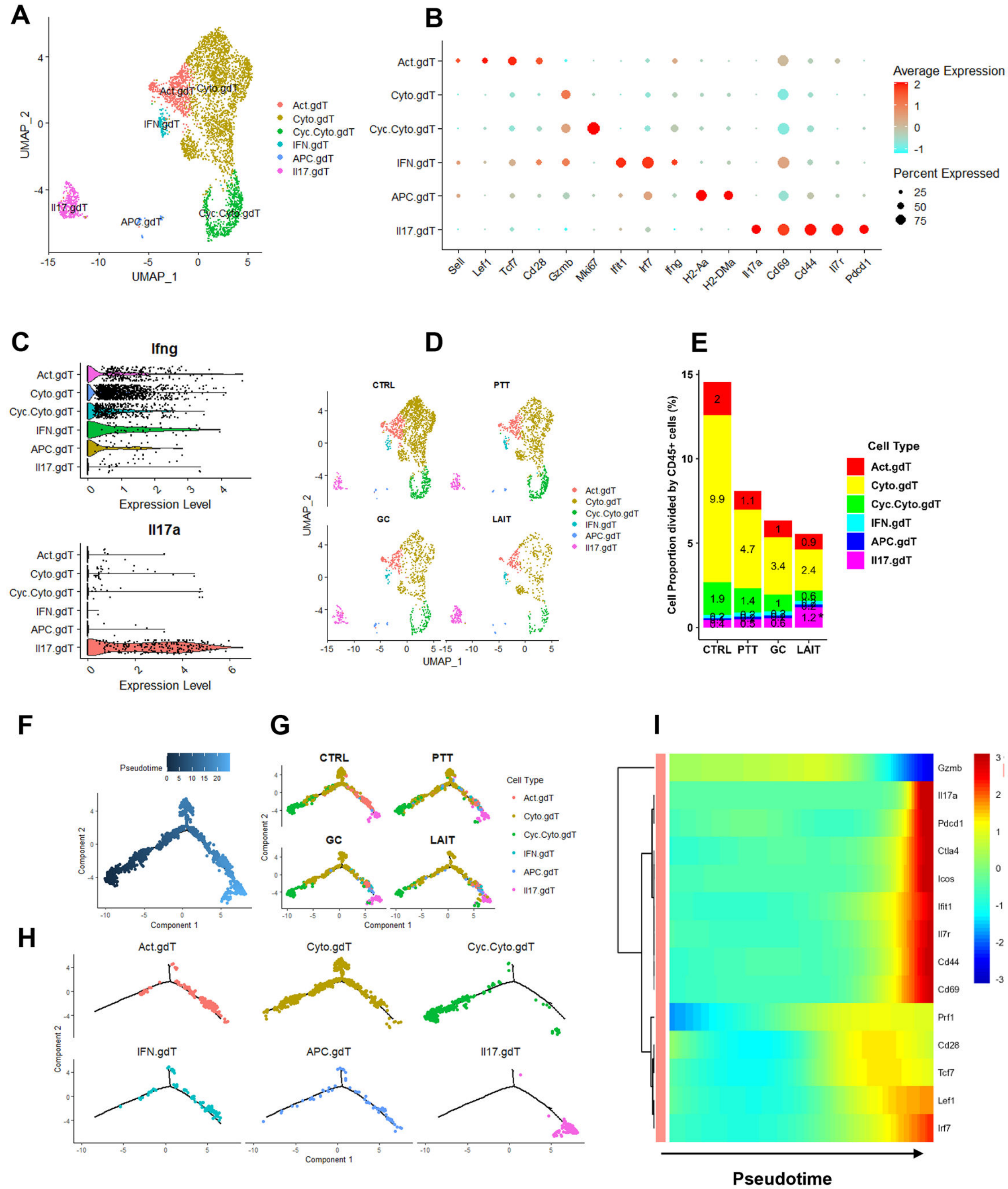

Figure 1. Analysis of tumor-infiltrating γδT cells in different treatment groups (CTRL, PTT, GC, and LAIT).

(A) Two-dimensional visualization of single-cell clusters using the method of Uniform Manifold Approximation and Projection (UMAP) for γδT cells from all treatment groups. γδT cells were classified into activated (also as central memory/CM, *Sell*⁺ *Cd44*⁺ *Il7r*⁺), cytotoxic (*Gzmb*⁺), cycling cytotoxic (*Mki67*⁺ *Gzmb*⁺), IFN (*Ifng*⁺ *Irf7*⁺), APC (*H2-Aa*⁺), and Il17 (*Il17a*⁺) γδT cells.

(B) Dot plot showing the expression of selected markers for γδT cell subtypes.

(C) Violin plots showing complementary expression patterns of *Ifng* and *Il17a* in γδT cell subtypes.

(D) UMAP plots of γδT cells in different treatment groups (CTRL, PTT, GC, and LAIT).

(E) Bar plots showing the γδT cell proportion in each cluster in different treatment groups (CTRL, PTT, GC, and LAIT). The proportion was calculated by using γδT cell number in each cluster divided by the total $CD45^+$ immune cells within each treatment. * indicates the statistical significance with $P<0.05$ from comparing LAIT with CTRL.

(F) Cell trajectory plot of γδT cells colored by pseudotime.

(G) Cell trajectory plots of γδT cells colored by subtypes in different treatment groups (CTRL, PTT, GC, and LAIT).

(H) Cell trajectory plots of each γδT cell subtype from all treatment groups.

(I) Heatmap showing expressions of selected γδT cell marker genes that significantly changed following pseudotime progression.

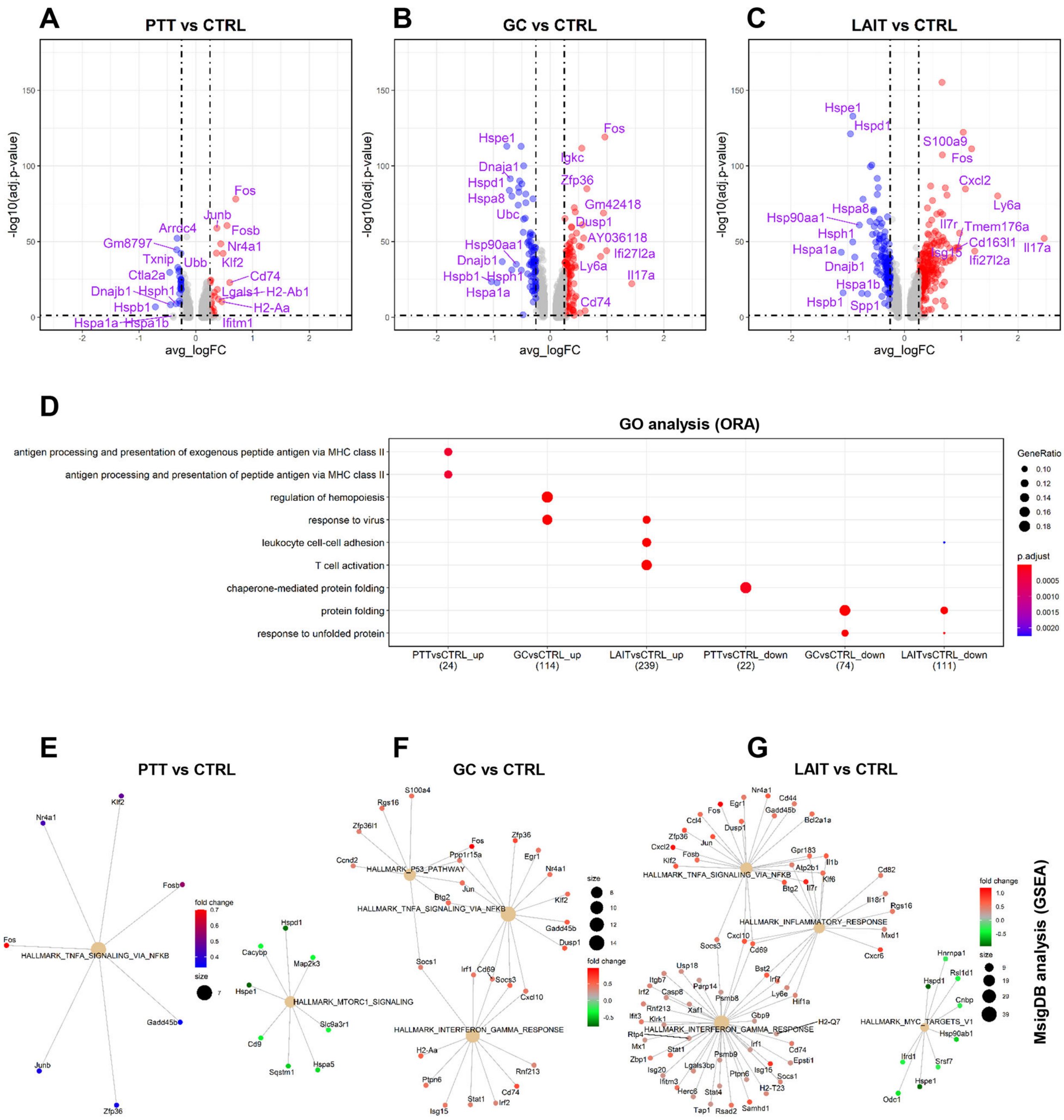

Figure 2. Differential gene expression and pathway enrichment analyses for tumor-infiltrating γδT cells.

(A) Volcano plot showing differential gene expression comparing PTT with CTRL (PTT vs CTRL). Top 10 upregulated and downregulated genes were labeled.

(B) Volcano plot showing differential gene expression comparing GC with CTRL. Top 10 upregulated and downregulated genes were labeled.

(C) Volcano plot showing differential gene expression comparing LAIT with CTRL. Top 10 upregulated and downregulated genes were labeled.

(D) Dot plot showing biological process (BP) of gene ontology (GO) analysis using over-representation analysis (ORA) method for both upregulated (up) and downregulated (down) genes for PTT vs CTRL, GC vs CTRL, and LAIT vs CTRL.
(E) Network plot showing gene set enrichment analysis (GSEA) for DEGs from PTT vs CTRL using MsigDB hallmark gene sets.
(F) Network plot showing the GSEA for DEGs from GC vs CTRL.
(G) Network plot showing the GSEA for DEGs from LAIT vs CTRL.

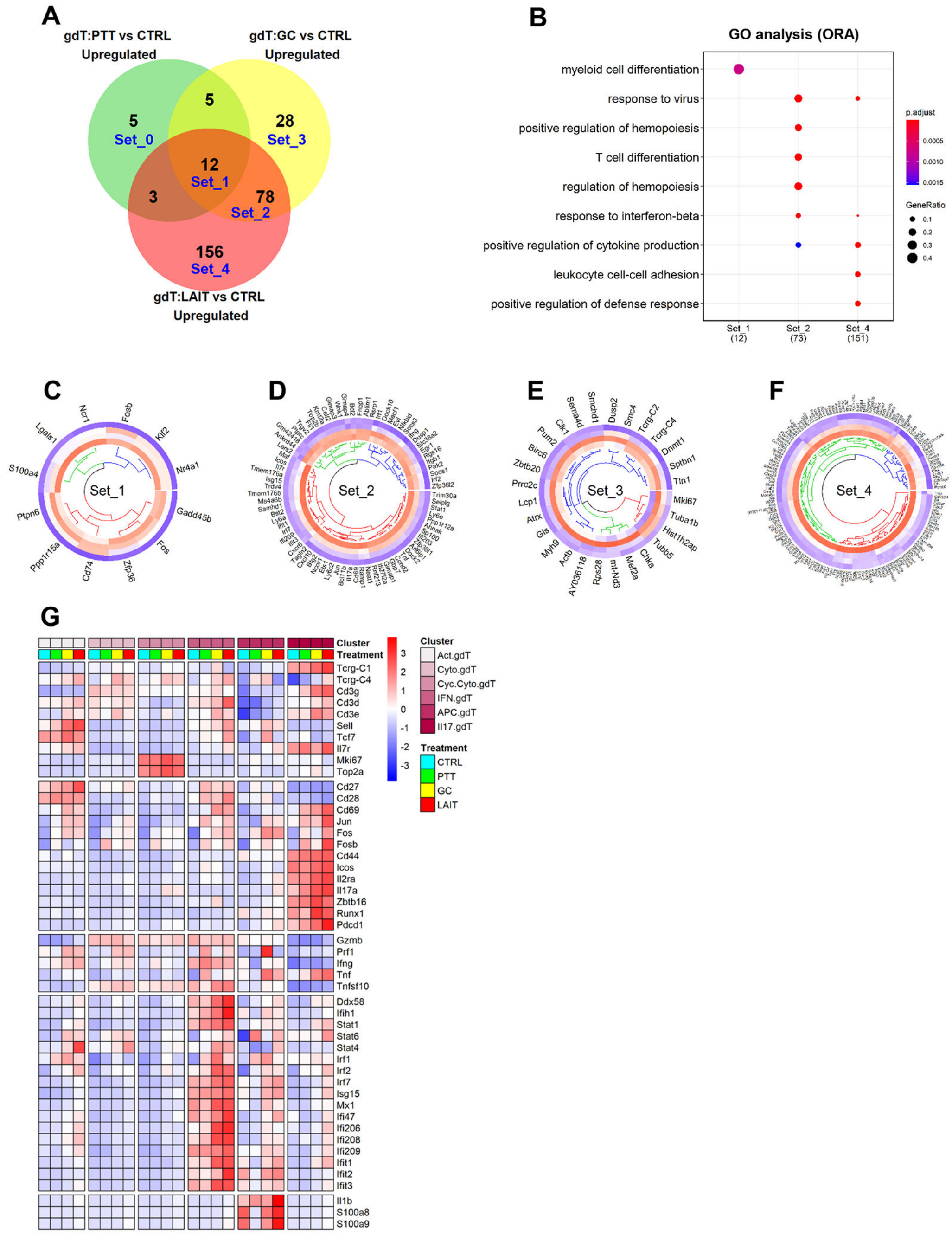

Figure 3. Overlap analysis of treatment-upregulated genes.

(A) Venn diagram showing upregulated genes from comparisons of PTT vs CTRL, GC vs CTRL, and LAIT vs CTRL. Five gene sets, from Set_0 to Set_4, are labeled.

(B) Dot plot for BP of GO analysis of genes in Set_0 to Set_4 (Enrichment of Set_0 or Set_3 not found).

(C) Circular heatmap showing the expression of genes from upregulated set 1 (Set_1) in each treatment group. Heatmap columns for groups of CTRL, PTT, GC, and LAIT were arranged from outside to inside. Higher expression was colored in red while lower in blue.

Author Manuscript

Author Manuscript

Author Manuscript

(D) Circular heatmap showing the expression of genes from upregulated set 2 (Set_2) in each treatment group.

(E) Circular heatmap showing the expression of genes from upregulated set 3 (Set_3) in each treatment group.

(F) Circular heatmap showing the expression of genes from upregulated set 4 (Set_4) in each treatment group.

(G) Heatmap showing the expression of selected genes in each treatment group in each γδT cell subtype.

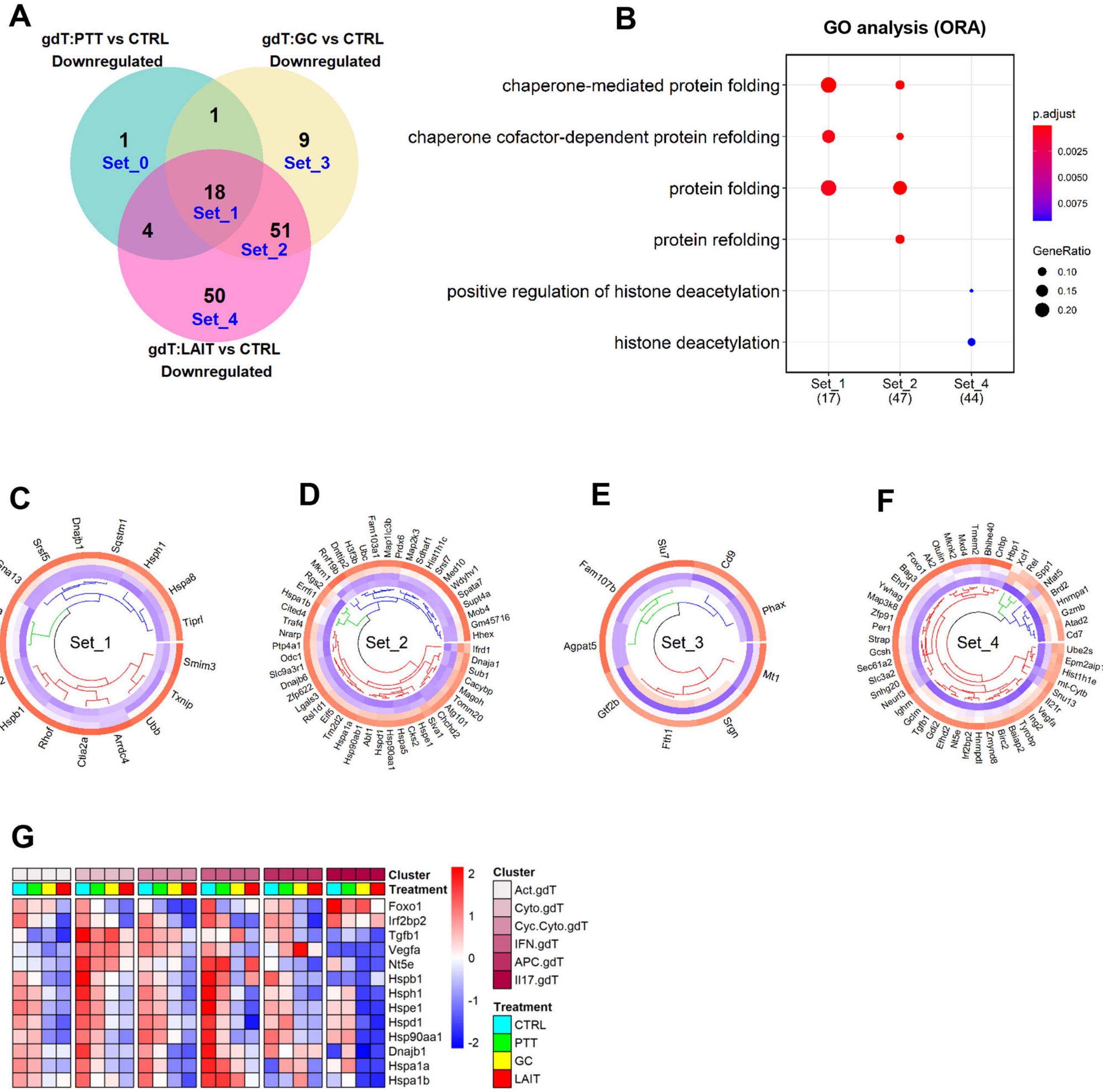

Figure 4. Overlap analysis of treatment-downregulated genes.

(A) Venn diagram showing downregulated genes from comparisons of PTT vs CTRL, GC vs CTRL and LAIT vs CTRL. Five gene sets, from Set_0 to Set_4, were labeled.

(B) Dot plot for BP of GO analysis of genes in Set_0 to Set_4 (enrichment of Set_0 or Set_3 not found).

(C) Circular heatmap showing the expression of genes from downregulated set 1 (Set_1) in each treatment group. Heatmap columns for groups of CTRL, PTT, GC, and LAIT were arranged from outside to inside. Higher expression was colored in red while lower in blue.

(D) Circular heatmap showing the expression of genes from downregulated set 2 (Set_2) in each treatment group.

(E) Circular heatmap showing the expression of genes from downregulated set 3 (Set_3) in each treatment group.

(F) Circular heatmap showing the expression of genes from downregulated set 4 (Set_4) in each treatment group.

(G) Heatmap showing the expression of selected immune suppressive genes in each treatment group in each γδT cell subtype.

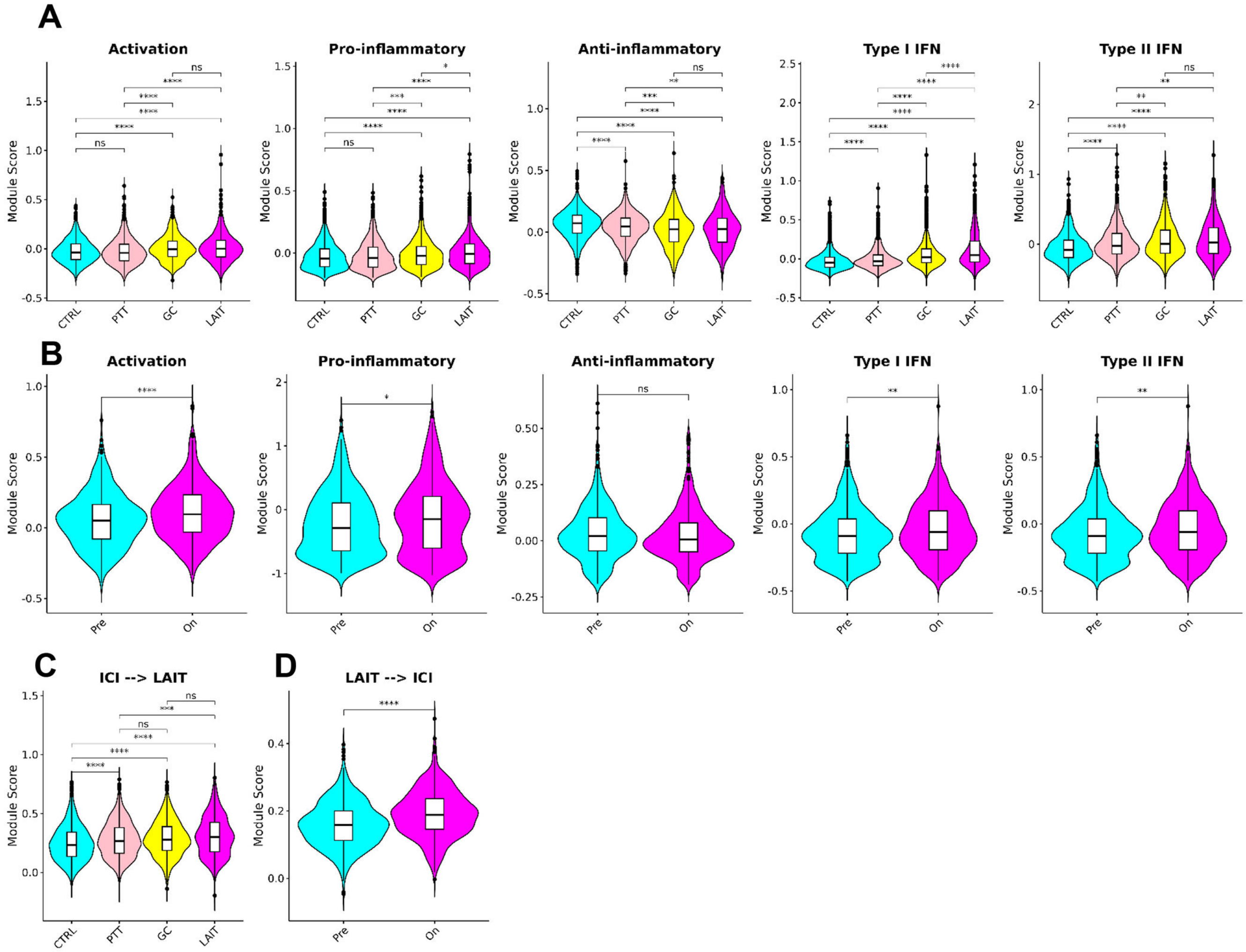

Figure 5. Correlation of LAIT and ICI stimulated antitumor phenotype of γδT cells.

(A) Violin plots showing module scores for selected gene signatures of γδT cells from mice bearing PyMT breast tumors in different treatment groups (CTRL, PTT, GC, and LAIT). Antitumor gene signatures, such as activation, pro-inflammatory, type I interferon, and type II interferon are enhanced by different treatments, especially LAIT. Pro-tumor gene signature of anti-inflammatory is reduced by the different treatments.

(B) Violin plots showing module scores for selected gene signatures of γδT cells from breast cancer patients before (*Pre*) and after (*On*) ICI treatment (Bassez et al., 2021). ICI induced antitumor gene signatures include activation, pro-inflammatory, type I interferon, and type II interferon, while ICI-inhibitted pro-tumor gene signature includes anti-inflammatory.

(C) Violin plots showing module scores of the gene signatures of γδT cells from mice bearing PyMT tumors after different treatments (CTRL, PTT, GC, and LAIT), focusing on the upregulated genes in TINK cells from ICI-treated breast cancer patients (Bassez et al., 2021).

(D) Violin plots showing module scores of gene signatures of γδT cells from breast cancer patients after ICI treatment (Bassez et al., 2021), focusing on the upregulated genes in γδT cells from PyMT tumors after LAIT treatment. The bilateral correlations of LAIT and ICI upregulated gene signatures in TINK cells from mouse breast tumors and patent breast tumors demonstrated the common effect of ICI and LAIT in stimulating γδT cells for antitumor immunity.

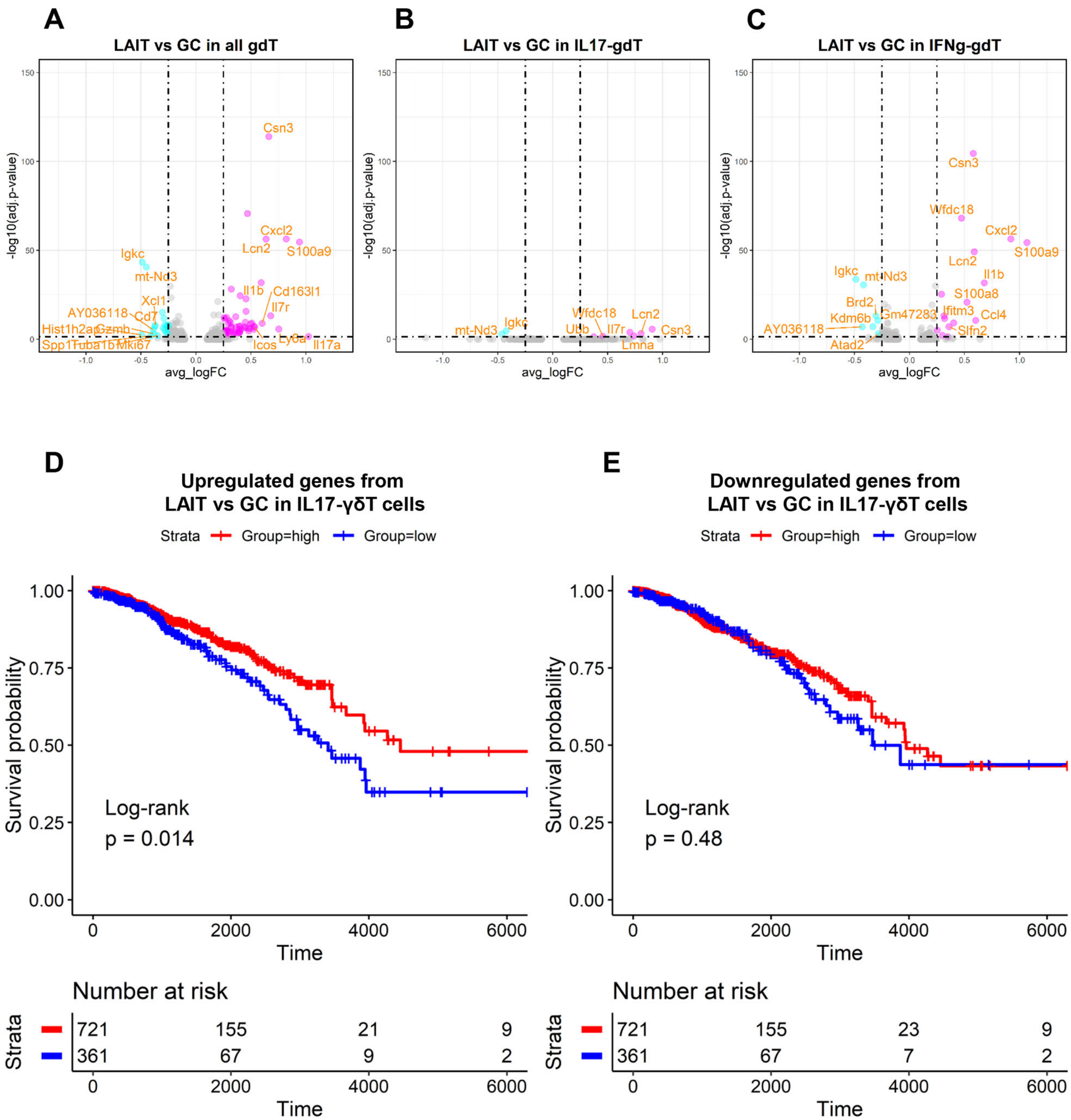

Figure 6. Analysis of the potential association of LAIT specifically upregulated genes with the survival of patients with breast cancer.

(A) Volcano plot showing differential gene expression comparing LAIT with GC in all γδT cells. Top 10 upregulated and downregulated genes are labeled.

(B) Volcano plot showing differential gene expression comparing LAIT with GC in IL17-γδT cells. Top 10 upregulated and downregulated genes are labeled.

(C) Volcano plot showing differential gene expression comparing LAIT with GC in IFNg-γδT cells. Top 10 upregulated and downregulated genes are labeled.

(D) Kaplan-Meier plots showing the significant difference in survival time (days) between breast cancer patients in groups with “high” and “low” expressions of LAIT vs GC-derived upregulated genes in IL17-γδT cells. Patient groups were stratified by the median of enrichment score calculated by gene set variation analysis (GSVA). The log-rank method was used for statistical analysis.

(E) Kaplan-Meier plots showing the insignificant difference in survival time (days) between breast cancer patients in groups with “high” and “low” expressions of LAIT vs GC-derived downregulated genes in IL17-γδT cells. Patient groups were stratified by the median of enrichment score calculated by gene set variation analysis (GSVA). The log-rank method was used for statistical analysis.